Enhanced performance monitoring architecture for synchronous digital hierarchy networks

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Existing high-speed network performance monitoring and measurement devices lack the flexibility to accommodate a comprehensive set of rapidly changing network conditions that require real-time information to ensure network integrity and synchronisation. This paper presents a new enhanced performance measurement/monitoring architecture (EPMA) that provides flexible, centralised, and distributed real-time information monitoring, analysis, and interpretation to support synchronous digital hierarchy (SDH) network management. The paper documents the use of existing SDH interfaces, test tools, and network analysis devices to support EPMA, and presents ideas for modifying existing SDH network equipment to provide EPMA functionality. Also presented are results of a feasibility demonstration of the basic EPMA functions using both existing and EPMA-customised SDH equipment on a model SDH network. These results demonstrate the utility of the EPMA.

1. Introduction

No existing high-speed telecommunicaions network management system (NMS), network performance measurement and monitoring device, or network test device individually includes the flexibility to provide a unified method for monitoring and managing a comprehensive set of rapidly changing network conditions that require real-time information to ensure network integrity and synchronisation.

This paper proposes a technique, called the enhanced performance monitoring architecture (EPMA), that combines these many features into a single tool for synchronous digital hierarchy (SDH) networks. The SDH EPMA builds upon a previous application of the EPMA technique to LANs [1].

The paper also documents the use of existing SDH interfaces, test tools, and network analysis devices on a model synchronous transport module-1 (STM-1) SDH network to provide a prototype implementation of EPMA. Also presented are ideas for modifying existing SDH network equipment to provide additional EPMA functionality, and results from a demonstration of basic EPMA functions using both existing and EPMA-customised SDH equipment and the model SDH network. This feasibility demonstration took place in March 1996 at BT Laboratories (BTL).

EPMA will provide flexible, centralised, and distributed network performance measurement and monitoring for real-time information that can be analysed and interpreted to support management of high-speed, multi-protocol networks. The key EPMA features include:

- real-time information collection and interpretation,
- flexibility, provided by dynamically reconfigurable hardware and software, to accommodate changing network conditions, multiple communications protocols, multiple customer premises equipment (CPE), and multiple network functions simultaneously,
- economical implementation via a single workstation or laptop PC that could additionally serve as a remote monitoring facility.

The ability to monitor real-time information requires observation of every bit of data that passes over the communications medium. The EPMA accomplishes such monitoring; however, the EPMA collects only certain pieces of this data that represent ‘information.’ Information refers to those portions of the data that are required as parameters for a particular network service or function, such as network synchronisation accuracy assessment, network error-performance monitoring, or network utilisation measurement. Because the EPMA collects only information rather than all of the data, memory and bandwidth requirements are low for an EPMA implementation.

The EPMA does more than merely collect the relevant information desired by the network functions or services —

it also interprets this information using the processing capability within a PC or workstation. The interpretation process need not be accomplished in real time depending upon the particular network management requirements. The complexity of the interpretation system can range from a simple observation of a busy or idle network state, with simple suggestions as to how the network operator should react, to a full expert interpretation system.

Flexibility represents the major distinguishing feature of the EPMA over existing measurement and monitoring tools. The EPMA uses dynamically reconfigurable hardware and software to enable reaction to changing network conditions. Hardware for the EPMA includes field programmable gate arrays, electrically erasable programmable read only memories (EEPROMs), and random access memories (RAMs). Using these devices along with a finite state machine algorithm [1, 2] provides an architecture that combines the higher speed of hardware with the flexibility of software. The software for the EPMA includes both the information interpretation software and hardware configuration files. If network conditions change during a measurement time interval, then the information interpretation system determines if a different set of parameters should be monitored and then uses a new configuration file residing in the workstation to reconfigure appropriately the EPMA hardware. Furthermore, this reconfiguration can take place without stopping the information collection process [1, 2].

Another distinct advantage of using dynamically reconfigurable hardware and software is that the EPMA can provide a subset of the network monitoring and measurement functions desired by a specific operator for a specific purpose. If the network operator wants to look at a different subset of functions or a different network protocol, then the EPMA can be reconfigured to accommodate these new requirements. The dynamic reconfigurability of the EPMA also permits the introduction of new performance measures that may be required or desired for future networks. For example, the EPMA feasibility demonstration introduces a new utilisation measure for virtual container (VC) — 12 tributaries that network operators can use for capacity planning purposes.

By not requiring simultaneous implementation of all functions that could possibly be required now and in the future, users can implement the EPMA central monitoring facility economically and use it for virtually any network. For the EPMA feasibility demonstration, the central monitoring facility comprised a single laptop computer, costing less than £1000 and having three serial interface ports. This facility monitored three aspects of SDH network performance, namely, SDH network utilisation, SDH error performance and alarms, and SDH network synchronisation accuracy. The central monitoring facility used the available information provided by the off-the-shelf SDH network elements whenever possible. For the network utilisation function, the network element incorporated software changes, based on BT's suggestions, that provided appropriate data outputs for the utilisation interpretation function.

The EPMA is also a distributed architecture. For example, a network specific EPMA can be implemented for each separate protocol in a multi-protocol network. A master EPMA can then monitor each of the network-specific EPMA implementations. If the master EPMA detects a problem with any of the network-specific EPMAs, then the information interpretation system of the master EPMA could initiate a reconfiguration of master EPMA so that it will monitor the same parameters as the EPMA that detected that network problem. Figure 1 shows a possible realisation for the EPMA that includes monitoring multiple communications protocol networks including SDH, SONET, Token Ring, Ethernet, Frame Relay, TCP/IP, Switched Multi-megabit Data Service (SMDS), and ATM. One version of the EPMA has already been implemented for token ring networks [1, 2]. Figure 1 also suggests that the distributed EPMA can monitor network synchronisation accuracy. For example, the SDH EPMA feasibility demonstration described in this paper monitored SDH network synchronisation accuracy based on global positioning system (GPS) satellite signals.

The remainder of this paper is organised as follows. Section 2 describes a feasibility demonstration of the basic EPMA functionality that uses both off-the-shelf and modified SDH equipment. Section 3 presents and interprets the results of this demonstration, and section 4 summarises the study findings and suggests extensions of and additional applications for this work.

2. EPMA feasibility demonstration for SDH networks

In March 1996, the basic EPMA functionality using SDH equipment and the model SDH network was demonstrated at BT Laboratories. The objectives of the SDH EPMA feasibility demonstration were to assess a new SDH network monitoring technique based on EPMA, to investigate the use of existing SDH interfaces, test tools, and network analysis devices to demonstrate the feasibility of the basic EPMA functions, to suggest and implement modifications to existing SDH network equipment so as to provide basic EPMA functionality, and to evaluate future development options for EPMA. The SDH EPMA feasibility demonstration accomplished these objectives through three tests — synchronisation accuracy monitoring, error performance and alarm monitoring, and utilisation computation and pattern detection at the VC-12 tributary level.

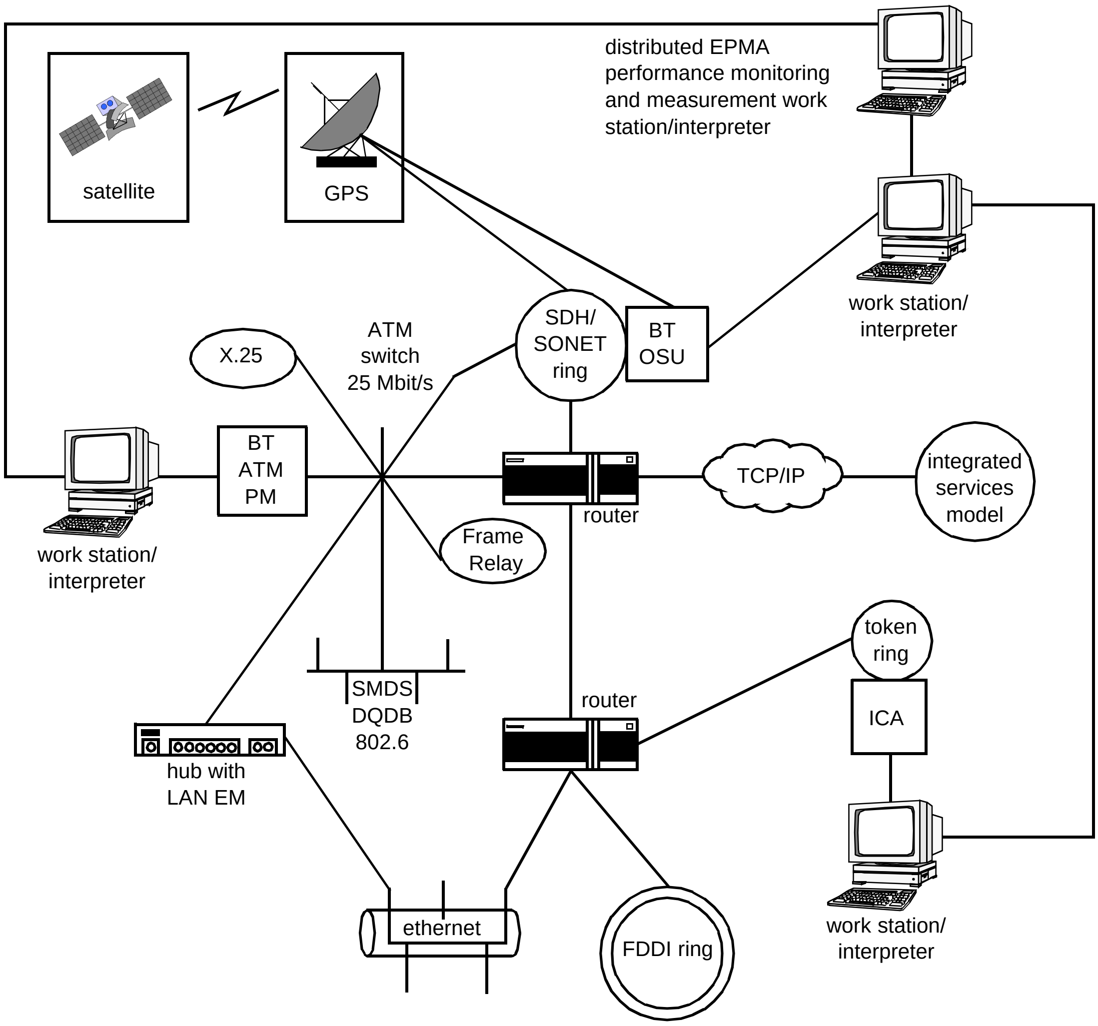

Fig 1 Distributed and centralised EPMA for performance measurement and monitoring.

2.1 *Demonstration set-up*

Figure 2 shows the laboratory set-up for the SDH EPMA feasibility demonstration.

The model SDH network is a ring with three ECI add/drop multiplexers (ADMs) that transmit traffic at STM-1 rates (i.e. 155 Mbit/s). The timing source for this ring is a 2.048 MHz clock signal from the Telecom Solutions Synchronous Supply Unit (SSU). During the SDH EPMA feasibility demonstration, a 1 MHz signal from this same clocking source was sent to the TrueTime XL-DC-602 GPS satellite receiver to observe fractional frequency offset and relative phase difference of this signal compared with a reference signal from the GPS receiver. A 1 pulse per second (PPS) signal from an HP 5071A caesium primary reference source [3] was also sent to this GPS receiver for the purpose of qualifying the accuracy of the GPS reference signal. These comparisons together provided a measure of the synchronisation accuracy for the SDH model network. Both the SSU and the GPS receivers obtained their signals from antennas residing at BTL.

The information gathered during all SDH EPMA feasibility demonstration tests was passed to the central monitoring facility via an RS-232 connection. This central monitoring facility includes the capability to simultaneously monitor and display multiple functions. For the SDH EPMA feasibility demonstration, the central monitoring facility was housed within a 75 MHz, 80486 DX4-based portable computer with an attached 21-inch colour monitor.

Also inserted within the model SDH ring was the performance and alarm (P&A) optical service unit (OSU) and the BT OSU. The P&A OSU is commercially available from GRCI under the trade name of OSU 155 [4, 5]. During the feasibility demonstration, the P&A OSU was surrounded by optical attenuators (namely, HP 157A Optical Attenuator 1300/1550 nm) on both the facility side (i.e. the side closest to the SDH ring) and the terminal side (i.e. the side closest to the terminal equipment). The optical attenuators were used to increase attenuation on the ring until alarms and performance errors were generated by the SDH equipment. The P&A OSU detected these errors and

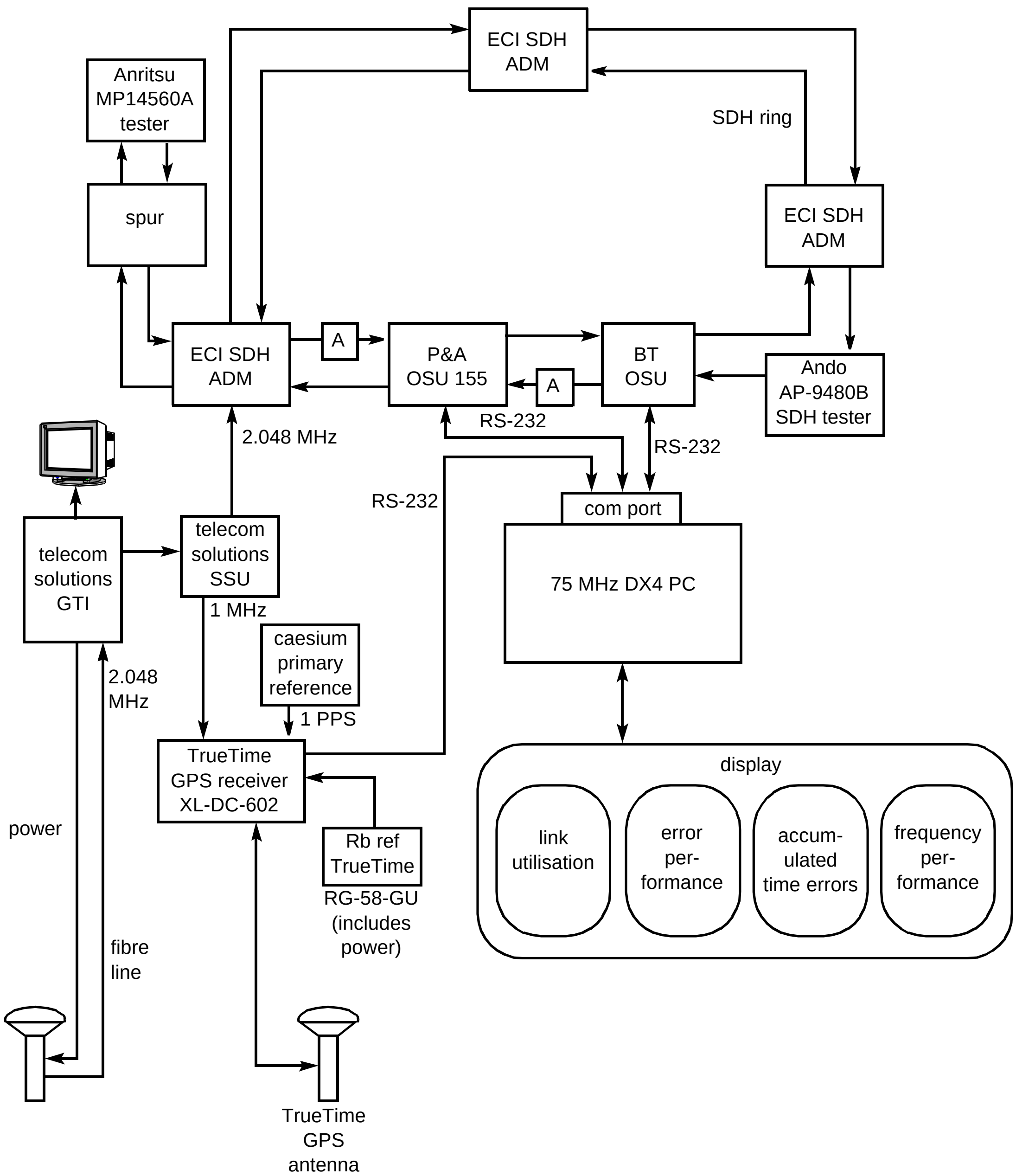

Fig 2 EPMA feasibility demonstration test set-up.

alarms and passed this information on to the central monitoring facility for display.

The BT OSU, also supplied by GRCI, is a modified version of the OSU 155 that detects specific byte patterns within a VC-12 payload for the purpose of pattern detection and SDH link utilisation computation. The modifications to the OSU 155 were made to enable EPMA functionality.

For the SDH EPMA feasibility demonstration, an ANDO AP-980B SDH tester and traffic generator was inserted within the SDH ring, between the BT OSU and one of the ECI ADMs. This tester passed programmed byte patterns on to the SDH ring and into the BT OSU to verify the BT OSU’s capability to detect the specific byte patterns on a VC-12 tributary.

An ANRITSU MP14560A tester and traffic generator was attached to a spur off an ECI ADM at a different location within the SDH ring.

This tester was used to verify that the BT OSU could detect patterns on specific VC-12 tributaries when these patterns were generated away from the BT OSU on the SDH ring and passed through other SDH network elements prior to observation and regeneration by the BT OSU. The information detected by the BT OSU was passed to the EPMA central monitoring facility for interpretation and display.

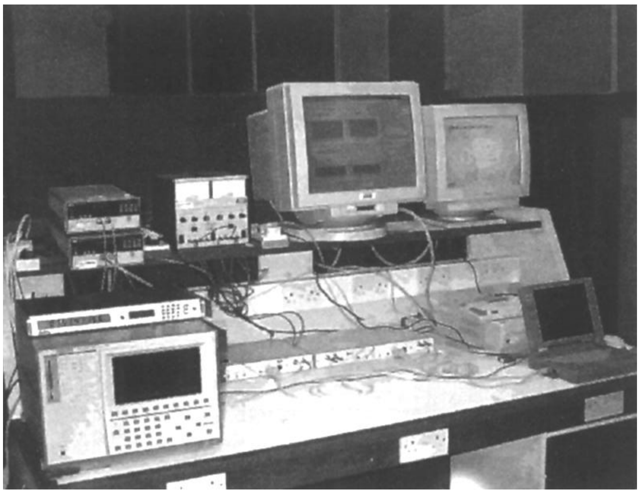

Fig 3 SDH EPMA feasibility demonstration laboratory set-up.

Figure 3 shows a photograph of the complete feasibility demonstration laboratory set-up.

2.2 SDH network synchronisation accuracy tests

The SDH network synchronisation accuracy tests used the TrueTime XL-DC-602 GPS receiver and TrueTime software to monitor frequency and time accuracy performance for the SDH network synchronisation source. This synchronisation source comprises a Telecom Solutions GPS Timing Interface (GTI) and a Telecom Solutions Synchronisation Service Unit (SSU). The TrueTime GPS receiver includes an external Rubidium source to provide a time comparison source. The TrueTime receiver used RG-58-GU cable to connect to the antenna receiving the GPS satellite signals.

The frequency accuracy portion of this test used a software program from TrueTime to compare the frequency offsets of an externally supplied 1 MHz signal with the signal from the rubidium disciplined GPS receiver. The external signal was taken from the Telecom Solutions SSU timing source for the model SDH network. The measurement interval was 1 s. The frequency resolution of the measurements was 6×10^{-11} Hz at one second averaging [6]. Information identifying the difference between the applied signal and the reference signal was passed to the EPMA centralised monitoring facility over an RS-232 link. The central monitoring facility plotted this information in real time on the display.

The time accuracy qualification portion of this test used a software program, written by TrueTime, to compare a 1 PPS signal from a caesium primary reference source with a reference signal from the rubidium disciplined GPS receiver. The applied signal was permitted to have a minimum pulse width of 100 ns. The rising edge of the pulse was measured with respect to the co-ordinated universal time (UTC) to within 30 ns [6]. Information identifying the difference between the applied signal and the reference signal was passed to the EPMA centralised monitoring facility over an RS-232 link. The central monitoring facility plotted this information in real time on the display.

Another part of the synchronisation accuracy tests provided a satellite visibility test in which the number of GPS satellites and the identification of the satellites being simultaneously tracked were monitored from the central monitoring facility.

2.3 Error performance and alarm monitoring tests

The error performance and alarm monitoring tests demonstrated the ability to incorporate the monitoring of error performance parameters (in line with ITU-T G.826 [7]) and alarms into the EPMA. These tests used the standard GRC OSU 155 [4, 5], referred to here as the P&A OSU. Figure 2 showed the point in the model SDH ring at which the P&A OSU was inserted during the demonstration. The P&A OSU served as both a signal regenerator and a network monitor for the performance and alarm tests. These tests required the following equipment — one model SDH ring, one ANDO AP9480B SDH analyser, two HP 157A optical attenuators (1300/1550 nm), one 75-MHz 80486 DX4 laptop PC running Windows95, and one external, high-quality monitor for displaying results. For these tests, the central monitoring facility used a command interpreter, written in Transaction Language 1 (TL1), so that the laptop PC could talk with the P&A OSU. This command interpreter and the TL1 gateway resided on the PC and communicated with the OSU over an RS-232 connection.

The SDH EPMA feasibility demonstration monitored and reported the following performance parameters and alarms:

- LOS — loss of signal,
- CVS — regenerator section level coding violations,
- SEFS — regenerator section severely errored framing seconds,
- ESS — regenerator section errored seconds,
- SESS — regenerator section severely errored seconds,
- CVL — multiplex section coding violations,
- CVLF — multiplex section far end coding violations,
- CVP — path level coding violations.

The optical attenuators were used to increase attenuation on the SDH ring until these alarms and errors occurred.

An additional test included using the central monitoring facility to inject the following error conditions on to the model SDH ring — B1 error locations within the regenerator section overhead, and B2 error locations within the multiplex section overhead. The verification of this error generation was provided by the ANDO SDH analyser. This test demonstrated the feasibility of using the central monitoring facility to affect network conditions and to evolve toward use as a network management system, as well as use as a network monitoring facility.

2.4 *VC-12 tributary pattern detection and utilisation monitoring*

The VC-12 tributary monitoring tests demonstrated the ability of the EPMA to monitor an STM-1 155 Mbit/s link at the VC-12 tributary level. These tests observed up to eight different repeated byte patterns for eight VC-12 tributaries that map into a VC-4 payload envelope. These tests required that GRCI modify the software for its standard OSU 155. In this paper, this modified OSU is called the BT OSU. As shown in Fig 2, the BT OSU 155 resided between the P&A OSU and an ECI ADM in the model SDH network during the SDH EPMA feasibility demonstration. The ANDO AP-9480B SDH tester generated VC-12 tributary patterns at the input to the BT OSU. The BT OSU was connected to the laptop PC central monitoring facility via an RS-232 connection. The BT OSU served as both a regenerator on this network and as a monitor during these tests. The additional equipment required at BTL for these tests included two traffic generators (ANRITSU MP14560A tester and ANDO AP9480B SDH analyser), one 75 80486 MHz DX4 portable PC running Windows95, and one external, high- quality monitor for displaying results.

2.4.1 EPMA VC-12 pattern detection

The SDH EPMA feasibility demonstration used the BT OSU to observe five different binary bit-pattern values for eight different tributaries. These binary values are listed in Table 1. The numbers in parentheses following the tributary identification are the K (TUG-3), L (TUG-2), and M (Tributary Unit (TU)-12) values for the VC-12 as specified in International Telecommunications Union (ITU) G.709 [8].

These binary values represent byte values that were injected into bytes of the bulk tributary unit — 12 (TU-12) payload using traffic generators inserted into the model SDH ring. The BT OSU detected these patterns and then passed the pattern detection information to the central monitoring facility that resides within the laptop PC. The information interpretation system installed in the central monitoring facility used selected pattern counts to determine utilisation of the SDH network, and the graphical user interface (GUI) displayed this information on a five second update basis.

Table 1 VC-12 pattern values.

VC-12 Number	(K,L,M)	Pattern Values
1	(1,1,1)	1111 1111
4	(1,2,1)	1111 1111
7	(1,3,1)	1111 1111
10	(1,4,1)	1111 1111
13	(1,5,1)	0000 0000
16	(1,6,1)	1010 1010
19	(1,7,1)	0110 1110
21	(1,1,2)	1111 0000

2.4.2 EPMA VC-12 utilisation

The SDH EPMA feasibility demonstration introduced two new utilisation performance measures. The first measure was based on detection of all ones in the VC-12 tributary. The presence of all ones on a VC-12 tributary indicates either an alarm indication signal (AIS) or an equipped tributary over which no useful information is transmitted. For example, Austin and Tomasson [9], in monitoring a particular network in which a customer sent a message to another end user, found that the all ones condition indicated an equipped tributary; however, they also found that the end user was not using the circuit. This situation resulted in an equipped but idle tributary [9]. Gagllardl, Mogavero, and Panarotto [10] suggest that the most efficient method for monitoring an SDH network at the link level is to identify idle cells or tributaries. The second EPMA utilisation measure is based on the detection of an unequipped tributary as indicated by the V5 byte. If bits 5, 6, and 7 of the V5 byte are all 0, then the VC-12 tributary is unequipped, i.e. unused [8, 11].

For the utilisation computation of the SDH EPMA feasibility demonstration, the BT OSU monitored each of four VC-12 tributaries (namely, VC-12 #1, VC-12 # 4, VC-12 # 7, and VC-12 #10) independently, and maintained statistics on the occurrence of the following conditions.

- The total number of polling intervals — a polling interval represents a 125 ms measurement time interval over which the BT OSU collects the desired statistical information. During 125 ms, one thousand 125- μs SDH frames are transmitted over the SDH network. The BT OSU accumulates statistical information for 5 s prior to transmitting this information to the EPMA central monitoring facility for interpretation and display. During 5 s, forty 125-ms measurement time intervals transpire which corresponds to 4×10^4 SDH frames. A 5 s information collection measurement time interval was chosen for the SDH EPMA demonstration to ensure that the BT OSU had time to transmit all required statistical information to the central monitoring facility.

- The number of intervals that satisfied the ‘ALWAYS’ condition — the ‘ALWAYS’ condition is set for the polling interval if the pattern byte value is continuously detected in every byte of the monitored tributary payload for the duration of the polling interval.

- The number of intervals that satisfied the ‘ONCE’ condition — the ‘ONCE’ condition is set for the polling interval if the pattern byte value is detected at least once in its corresponding tributary payload over the duration of the polling interval.

- An indicator of whether the VC-12 tributary was equipped or unequipped based on the contents of the V5 byte — only the V5 bytes of the last 64 frames of the 5 s measurement time interval were examined because of limited buffer space in the BT OSU.

The information interpretation system of the EPMA used these statistics as parameters for new EPMA utilisation performance measures. The EPMA utilisation measures 1—3 (below) can be computed in real time. These performance measures were computed and displayed during the SDH EPMA feasibility demonstration. Section 3 presents results based on these new utilisation performance measures.

Depending on the regenerator or physical tap into the network, the real-time data may require averaging. For these cases, the EPMA also includes averaged utilisation measures for accuracy compensation. EPMA averaged utilisation measures are presented in measures 4 and 5.

The following notation is used for the measures:

- L denotes the number of measurement time intervals over which results are averaged,

- N denotes the total number of VC-12 tributaries to be monitored,

- m denotes the measurement time interval.

Utilisation Measure 1 — Ones Always or Once Conditions (real time)

Let U_T denote total per cent utilisation, i.e. the per cent utilisation for the total number of tributaries, N, to be monitored.

Let I_n denote the number of idle frames per measurement time interval m for VC-12 tributary n, where n = 1,...,N. (Recall from the above discussion that m = 5 results in 4×10^4 SDH frames.)

$$U_T = \left(1 - \frac{1}{N}\sum_{n=1}^{N} I_n\right) \times 100 \qquad \text{...... (1)}$$

For $N = 4$,

I_1 = Number VC-12 #1 Always / # Frames in m sec,

I_2 = Number VC-12 #4 Always / # Frames in m sec,

I_3 = Number VC-12 #7 Always / # Frames in m sec,

I_4 = Number VC-12 #10 Always / # Frames in m sec.

This utilisation measure also applies to the ONCE condition by replacing the statistics for the ALWAYS condition with those of the ONCE condition.

Utilisation Measure 2 — Ones Always or Once Conditions (real time)

Let U_1 denote per cent utilisation for VC-12 #1.

$$U_1 = (1 - I_1) \times 100 \qquad \text{...... (2)}$$

where I_1 = Number VC-12 #1 Always / # Frames in m sec.

This utilisation measure also applies to the ONCE condition by replacing the statistics for the ALWAYS condition with those of the ONCE condition.

Utilisation Measure 3 — Equipped Tributary Condition (real time)

Let U_{ET} denote total per cent utilisation based on unequipped tributaries, where an unequipped tributary represents an unused tributary [8].

Let I_{EN} denote the number of idle frames per measurement time interval m for VC-12 tributary n, where n = 1,...,N.

$$U_{ET} = \left(1 - \frac{1}{N}\sum_{n=1}^{N} I_{En}\right) \times 100 \qquad \text{...... (3)}$$

For $N = 4$,

I_{E1} = Number VC-12 #1 Unequipped / # Frames in *m* sec,

I_{E2} = Number VC-12 #4 Unequipped / # Frames in *m* sec,

I_{E3} = Number VC-12 #7 Unequipped / # Frames in *m* sec,

I_{E4} = Number VC-12 #10 Unequipped / # Frames in *m* sec.

Utilisation Measure 4 (accuracy compensation with delayed results based on ones pattern detection)

Let I_{AT} denote the total per cent of idle frames per measurement time interval *m* for all VC-12 tributaries during all measurement time intervals.

Let I_{Aji} denote the per cent of idle frames per measurement time interval *m* for VC-12 tributary *j* and measurement time interval number *i*, where $j = 1,...,N$, and $i = 1,...,L$.

$$I_{AT} = \frac{1}{N \times L} \left(\sum_{j=1}^{N} \sum_{i=1}^{L} I_{Aji} \right) \qquad ...... (4)$$

Let U_{AT} denote total per cent utilisation for all tributaries over all measurement time intervals.

$$U_{AT} = (1 - I_{AT}) \times 100 \qquad ...... (5)$$

Let U_{A1} denote per cent utilisation for VC-12 tributary 1 over all measurement time intervals.

$$U_{A1} = \left(1 - \frac{1}{L} \sum_{i=1}^{L} I_{A1i} \right) \times 100 \qquad ...... (6)$$

Utilisation Measure 5 (accuracy compensation with delayed results based on unequipped tributary pattern detection)

Let I_{AET} denote the total per cent of unequipped (i.e. idle) frames per measurement time interval *m* for all VC-12 tributaries during all measurement time intervals.

Let I_{AEji} denote the per cent of unequipped (i.e. idle) frames per measurement time interval *m* for VC-12 tributary *j* and measurement time interval number *i*, where $j = 1,...,N$, and $i = 1,...,L$.

$$I_{AET} = \frac{1}{N \times L} \sum_{j=1}^{N} \sum_{i=1}^{L} I_{AEji} \qquad ...... (7)$$

Let U_{AET} denote total per cent utilisation for all tributaries over all measurement time intervals based on unequipped tributaries.

$$U_{AET} = (1 - I_{AET}) \times 100 \qquad ...... (8)$$

Let U_{AE1} denote per cent utilisation for VC-12 tributary 1 over all measurement time intervals based on unequipped tributaries.

$$U_{AE1} = \left(1 - \frac{1}{L} \sum_{i=1}^{L} I_{AE1i} \right) \times 100 \qquad ...... (9)$$

2.5 EPMA central monitoring facility

A distinguishing feature for the EPMA is its central monitoring facility. Even though many existing performance measurement and monitoring tools can monitor several aspects of a network, the EPMA improves these capabilities by providing simultaneous monitoring of multiple functions in one central unit.

2.5.1 Graphical user interface

Figure 4 shows the EPMA graphical user interface (GUI) with which the user can name the COMM port over which information is transmitted to the EPMA workstation, pick the COMM port speed, select a directory for EPMA log file storage, and view the log file.

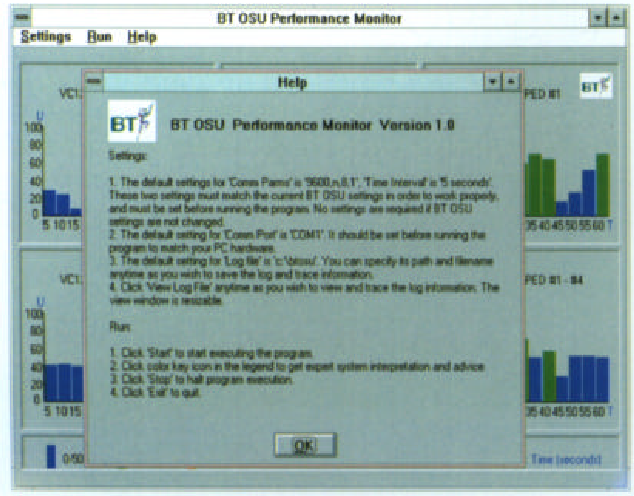

Fig 4 EPMA graphical user interface (GUI).

- EPMA utilisation and pattern displays

After set-up, the SDH EPMA GUI displays real-time utilisation information based on the ALWAYS condition, ONCE condition, and EQUIPPED TRIBUTARY condition statistics. The horizontal axis of each graph identifies the measurement time interval (MTI) over which the statistics are collected in seconds. In this case the overall time is 1 min (60 s) and each bar in the graph corresponds to an MTI of m = 5 s. The vertical axis represents per cent utilisation according to the appropriate performance measures in equations (1) — (9).

Figure 5 provides an example of this display. The three graphs at the top of this display represent utilisation computation for VC-12 # 1 only. The bottom three graphs represent average utilisation computed for VC-12 #1, VC-12 #4, VC-12 # 7, and VC-12 #10.

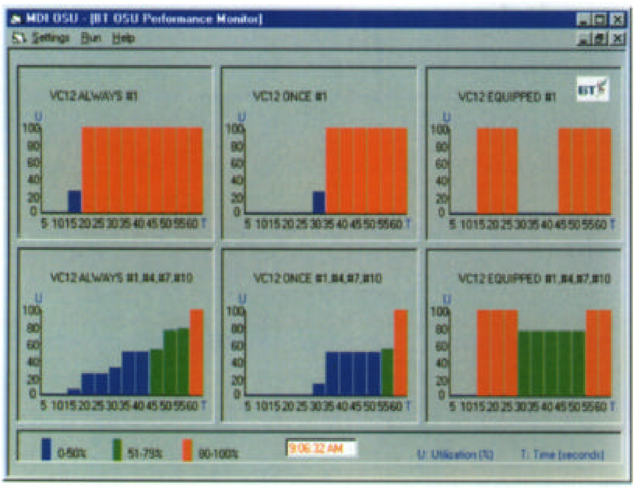

Fig 5 EPMA utilisation display.

The SDH EPMA GUI also displays pattern detection information. Figures 13 and 14 in section 3.3 provide examples of this display for different byte patterns detected during the SDH EPMA feasibility demonstration.

- EPMA trace facility

The SDH EPMA GUI also provides a trace facility that records in a log file all information passed to the EPMA information interpretation system from the SDH network through the RS-232 COMM port. Users can display this log file at any time. The trace information corresponds directly to the graphical information on the screen. Figure 12 in section 3.3 provides a graphical example of the EMPA trace facility for the SDH EPMA feasibility demonstration utilisation monitor. A similar trace facility exists for the EPMA pattern detection monitor.

2.5.2 Information interpretation system

The SDH EPMA provides an information interpretation system for network utilisation conditions. The utilisation display shows three colors, blue, green, and red, that correspond to a low utilisation state, a satisfactory utilisation state, and a high utilisation state, respectively. For example, Fig 6 displays a screen image for the high utilisation state. The interpretation system graphical display describes the satisfactory utilisation state as one in which the network bandwidth (i.e. capacity) properly matches user demand, and no action is required. A low utilisation state is one in which the available network capacity (i.e. bandwidth) greatly exceeds the user demand. The interpretation system provides two recommendations:

- if not peak traffic time, then no action is required,
- if peak traffic time, then reallocate bandwidth to other users if possible.

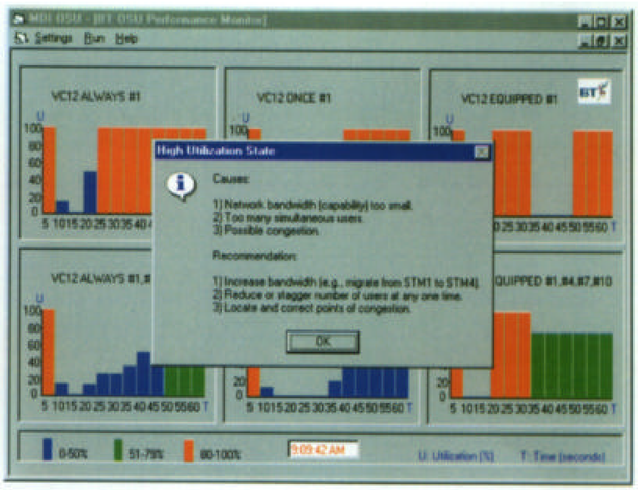

Fig 6 EPMA information interpretation system for high utilisation.

3. Results and observations

This section presents results obtained from the SDH EPMA feasibility demonstration. During this demonstration, the laptop PC provided a multi-function, central monitoring facility that simultaneously monitored SDH synchronisation accuracy, SDH network error performance, and SDH network utilisation.

3.1 SDH network synchronisation accuracy tests

The SDH network synchronisation accuracy tests used the TrueTime XL-DC-602 GPS receiver and TrueTime software to monitor frequency and time accuracy performance for the SDH network synchronisation source. Figures 7 and 8 show actual data as they appeared on the central monitoring facility during the frequency accuracy measurement test for fractional frequency offset and relative phase, respectively. The vertical axis of the fractional frequency offset graph represents the difference in the frequency of the 1 MHz signal provided by the Telecom Solutions SSU compared with that of the rubidium disciplined TrueTime GPS receiver. The scale of the vertical axis is pico seconds/second (ps/s). The horizontal axis represents running time of the test in seconds.

Figure 7 shows that in this 198-second running time sample, the maximum fractional frequency offset between the Telecom Solutions SSU and the TrueTime rubidium disciplined GPS receiver was ± 126.5 ps/s.

Figure 8 shows that the relative phase difference between the Telecom Solutions SSU and the TrueTime rubidium disciplined GPS receiver ranged between + 0.126 ns and -0.253 ns in the same 198-second running time sample.

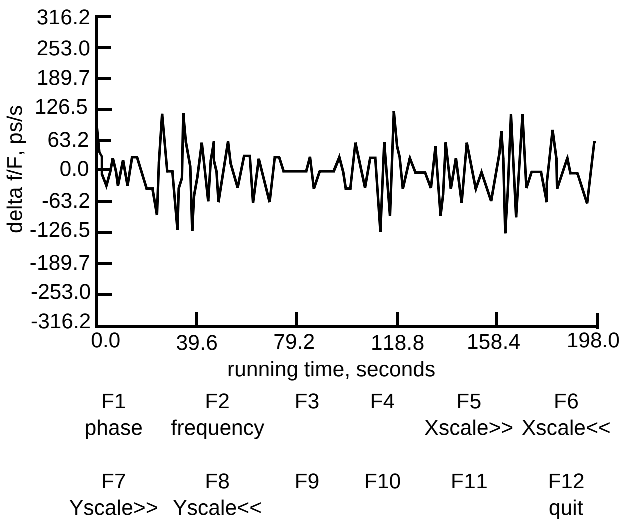

Fig 7 EPMA frequency measurement test fractional frequency offset (actual test data).

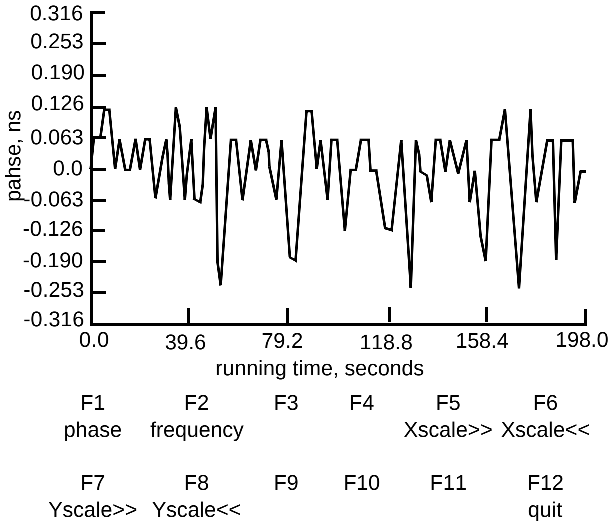

Fig 8 EPMA frequency measurement test relative phase (actual test data).

Figures 9 and 10 show actual sets of data as they appeared on the central monitoring facility during the time accuracy measurement test, described in section 2.2, for fractional frequency offset and relative phase, respectively. Figure 9 shows that during the approximately 7000-second running time, the peak fractional frequency offsets between the HP caesium primary reference source and the TrueTime rubidium disciplined GPS receiver were approximately $\pm$ 31 600 ps/s.

Figure 10 shows that the relative phase difference between the HP caesium primary reference source and the TrueTime rubidium disciplined GPS receiver ranged between + 0.0 nanoseconds (ns) and -150 ns during this same 7000-second running time sample. Note that steps in the relative phase plot in Fig 10 occur at the same running

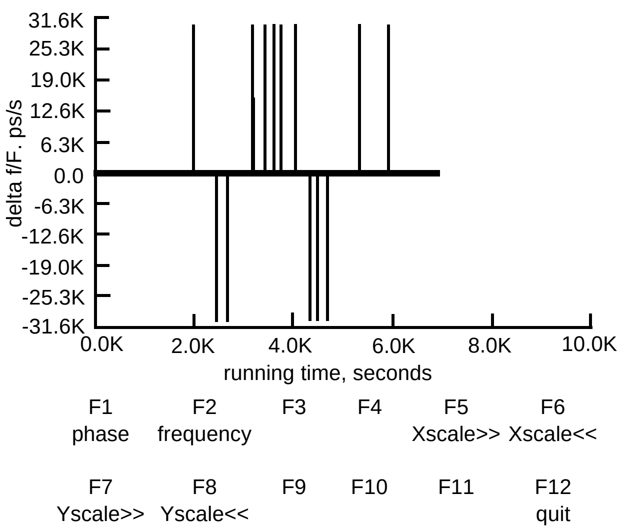

Fig 9 EPMA time accuracy test fractional frequency offset (actual test data).

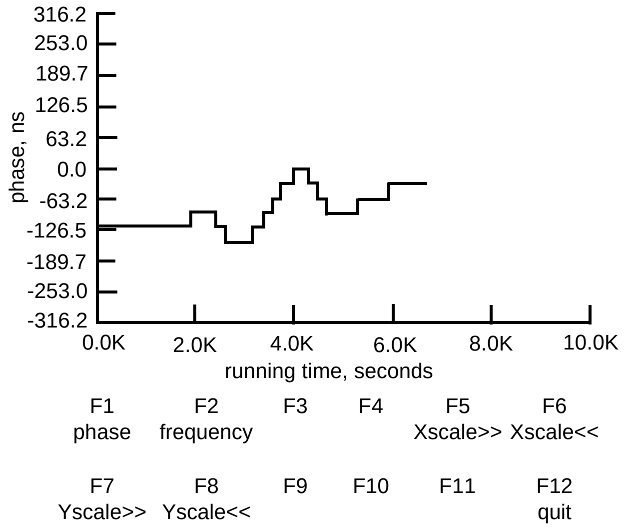

Fig 10 EPMA time accuracy measurement test relative phase (actual test data).

times and in the same directions as the spikes in the fractional frequency offset plot in Fig 9. These perturbations may be caused by the relative shifting of the GPS satellite positions with respect to the GPS antenna location at BTL.

On a separate screen the central monitoring facility identified the number of GPS satellites on to which the GPS receiver was presently locked, the identification numbers of these satellites, the state of the satellite (e.g. good health), whether the satellite could provide data that the GPS receiver could use (e.g. enabled), whether the satellite was presently being tracked by the GPS receiver, and the relative signal quality provided by the GPS receiver (by indicating a dimensionless number between 4 and 20 with the greater number representing a higher signal quality). For the SDH EPMA feasibility demonstration this screen indicated that the GPS receiver was locked on to six satellites including satellites 6 and 25, that these satellites were in good health, enabled, and presently tracked, and that the relative signal quality was good. (The GPS receiver provided a signal quality indication for satellite 6 of + 18.42 and for satellite 25 of + 12.98.)

3.2 Error performance parameter monitoring

The error performance parameter and alarm monitoring tests used existing SDH monitoring equipment and vendor-supplied software to interpret and display the error performance and alarm information on the EPMA central monitoring facility. These tests took advantage of the rich performance information in the SDH protocol.

Table 2 provides a summary of the actual error performance monitoring data captured and displayed during tests for both the facility side and the terminal side of the P&A OSU. Recall that these errors were generated by increasing attenuation on the SDH ring using optical attenuators on both sides of the P&A OSU. The first column of Table 2 identifies either the terminal (TRM) or the facility (FAC) side of the P&A OSU. The second column identifies the type of error that was monitored. The third column indicates the error count for the present MTI. The fourth column identifies a threshold that, if exceeded by the error count, will generate an alarm within the P&A OSU. The fifth column indicates the MTI over which the P&A OSU accumulates error count. The sixth column identifies the OSU register in which the error count information is stored.

Table 2 EPMA performance monitoring — facility and terminal sides (actual test data for 13 March 1996, 15:13:01).

P&A OSU	Error	Error Count	Thres-hold	MTI (min)	OSU Register
FAC	CVS	0	1146	15	0
FAC	SEFS	335	225	15	0
FAC	ESS	335	225	15	0
FAC	SESS	335	225	15	0
FAC	CVL	110	0	15	0
FAC	CVLFE	0	0	15	0
FAC	CVP	0	0	15	0
TRM	CVS	0	1146	15	0
TRM	SEFS	470	225	15	0
TRM	ESS	470	225	15	0
TRM	SESS	470	225	15	0
TRM	CVL	212	0	15	0
TRM	CVLFE	0	0	15	0
TRM	CVP	0	0	15	0

Table 3 provides a summary of the alarm data captured and displayed during the SDH EPMA feasibility demonstration tests. The abbreviations used in Table 3 are the same as those in Table 2 with the following additions — CR: critical alarm, MN: minor alarm, and MJ: major alarm. Column 1 of Table 3 identifies the P&A OSU side on which the alarm was detected (namely, FAC or TRM). Column 2 identifies the interface type (e.g. OC3 or STS-1). Column 3 identifies the alarm severity (i.e. CR, MJ, or MN). Column 4 identifies the alarm type. The P&A OSU detects and displays particular alarms after the appropriate error condition persists for a period greater than the threshold time interval that is identified in column 5. The notation for such an alarm is ‘T-error’.

Table 3 EPMA alarm monitoring (actual test data, 13 March 1996, 15:18:45).

P&A OSU side	Interface type	Alarm severity	Alarm type	Threshold time interval (minutes)
FAC	OC3	CR	LOS	15
TRM	OC3	CR	LOS	15
FAC	OC3	MN	T-ESS	15
TRM	OC3	MN	T-ESS	15
FAC	OC3	MN	T-SEFS	15
TRM	OC3	MN	T-SEFS	15
FAC	OC3	MN	T-SESS	15
TRM	OC3	MN	T-SESS	15

During the SDH EPMA feasibility demonstration, the following key strokes from the central monitoring facility instructed the P&A OSU to inject B1 and B2 errors on to the SDH ring:

‘ed-oc3::all:ctag:::errinj=b1104,errinj=b2104;’

A central monitoring facility screen displayed this error generation command.

3.3 *VC-12 tributary utilisation and pattern detection monitoring*

The VC-12 tributary utilisation and pattern detection monitoring tests successfully demonstrated real-time monitoring and interpretation for detected patterns. Figure 11 shows a screen of actual data captured during the utilisation test with the ANDO tester generating traffic on to the model SDH ring. Using equations (1) — (3), the information interpretation system computed utilisation for the ALWAYS, ONCE, and EQUIPPED TRIBUTARY conditions discussed in section 2.4.

The graphs in Fig 11 show twelve 5 s measurement time intervals, beginning at 5 s and ending at 60 s. During the first two measurement time intervals (i.e. 5 and 10), the utilisation measures all show the model SDH network utilisation to be 0%. Recall that utilisation of 0% for the ALWAYS condition means that all ones fill the entire VC-12 tributary payload during the measurement time interval, that utilisation of 0% for the ONCE condition means that at least one byte, but not all bytes, within the VC-12 payload is filled with all ones during the measurement time interval, and that utilisation of 0% for the EQUIPPED condition means that all of the last 64 VC-12 tributaries during the measurement time interval are unequipped.

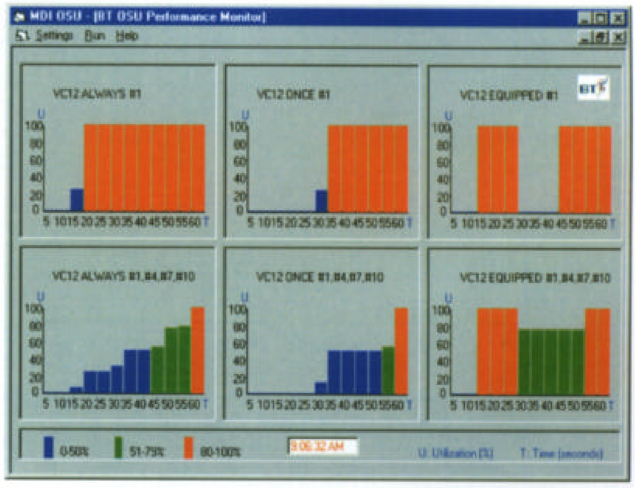

Fig 11 EPMA utilisation measurement (actual test data with ANDO traffic generator).

During the third measurement time interval (i.e. 15 s), VC-12 ALWAYS #1 utilisation is 25% and, by 20 s, VC-12 ALWAYS #1 utilisation equals 100% (i.e. other than all ones fills the payload of VC-12 #1). The average utilisation computation for VC-12 ALWAYS #1, #4, #7, #10 reports 25% average utilisation at 20 s caused by VC-12 ALWAYS #1 being 100% utilised and the other three tributaries remaining idle.

However, utilisation for the ONCE condition during the fourth measurement time interval (i.e. 20 s) still shows 0% utilisation. This could be caused by a single byte of all ones appearing in the VC-12 #1 payload. This example points out how crude the ONCE condition utilisation measure is. However, this measure serves as a useful tool during initial network set-up and debug or in the absence of pointer processing. Pointer processing uses bytes within the SDH overhead to determine the location of the byte where the VC-*n* begins, where *n* is the tributary level. For example, in the SDH EPMA feasibility demonstration, the BT OSU did not implement pointer processing at the VC-12 tributary level. The technology existed to accomplish such pointer processing, but this feature could not be implemented within the BT OSU in time for the demonstration. In order to verify that the EPMA indeed could detect the ALWAYS condition, the team used known traffic patterns and the ONCE condition to assist in lining up the pointers for the VC-12 tributaries. Using the ONCE condition for this purpose saved considerable time in setting up the demonstration.

During the fourth measurement time interval (i.e. 20 s), the VC-12 EQUIPPED #1 condition agreed with the VC-12 ALWAYS #1 condition, with both conditions indicating 100% utilisation for tributary VC-12 #1. For VC-12 #4, VC-12 #7, and VC-12 #10, the EQUIPPED tributary condition showed 100% equipped tributaries, and yet, the ALWAYS condition utilisation measure showed that the payloads of these tributaries were filled with all ones. This 'equipped tributaries with idle information condition' could represent a situation where, even though the customer has paid for the bandwidth of a VC-12 tributary, the customer is not transmitting useful information over this tributary. If such a condition were to persist for a long period of time, then the EPMA information interpretation system could identify excess capacity on the SDH network.

The EPMA log file is an effective capability for determining traffic activity trends. Figure 12 shows the log file that corresponds to the graphical utilisation information shown in Fig 11. The first number in the first line in Fig 12 (i.e. 40) represents the number of 125-ms polling intervals observed by the BT OSU, the next three numbers represent the number of polling intervals in which the 'ALWAYS', 'ONCE', and 'EQUIPPED tributaries' conditions, respectively, were detected by the BT OSU for VC-12 #1. The next 21 numbers on line 1 provide this same information for the remaining seven VC-12 tributaries that were monitored. The last number of line 1 is a time stamp. Line 2 provides the % utilisation for the ALWAYS #1, ONCE #1, and EQUIPPED #1 conditions. Line 3 provides the average % utilisation for the ALWAYS #1, #4, #7, #10, ONCE #1, #4, #7, #10, and EQUIPPED #1, #4, #7, #10 conditions. The central monitoring facility updates the log file at the end of each measurement time interval (i.e. 5 s for the SDH EPMA feasibility demonstration).

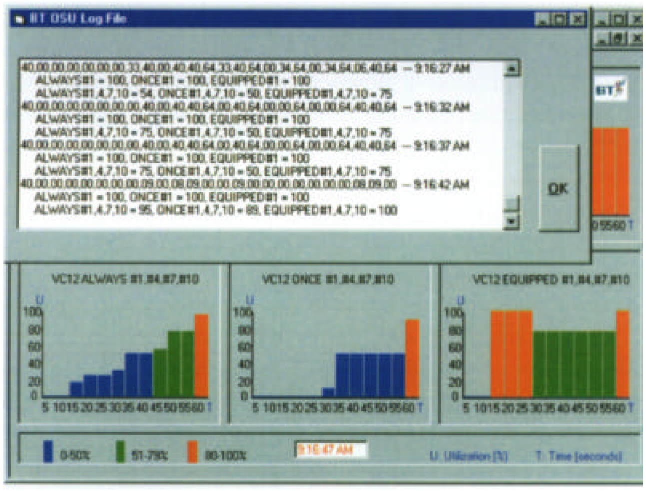

Fig 12 EPMA utilisation measurement log file (actual test data with ANDO traffic generator)

During measurement time intervals 5 to 12 (i.e. 25 s through 60 s), the ANDO traffic generator passed other than all ones traffic on to tributaries VC-12 #1, #4, #7, and #10, in turn, until the model SDH ring was 100% utilised for all four tributaries at measurement time interval 12 (i.e. 60 s). Notice that average utilisation sometimes fell between 0%, 25%, 50%, 75%, and 100% because the traffic generator increased traffic load in the middle of a measurement time interval.

The SDH EPMA feasibility demonstration also showed that the EPMA, using the BT OSU, could detect selected byte patterns on selected VC-12 tributaries. Figure 13 shows four different byte patterns on four different VC-12 tributaries that were detected during the SDH EPMA feasibility demonstration. The detected VC-12 byte patterns included: 00000000, 10101010, 01101110, and 11110000. These patterns were selected to show the diversity of the pattern detection capability. The vertical axis of each graph, along with the bar colours, identifies whether the particular pattern was detected once, always, or never, or if the data was bad.

Table 4 provides the bar colour interpretation key. The horizontal axis of each graph identifies the MTI over which the patterns were detected and collected. For the SDH EPMA feasibility demonstration the MTI equalled 5 s.

Table 4 Colour-coded key for EPMA pattern detection graph.

Bar colour	Meaning
Blue	Selected pattern fills every byte within the selected VC-12 tributary payload during the measurement time interval.
Green	Selected pattern fills at least one byte, but not all bytes, within the selected VC-12 tributary payload during the measurement time interval.
Red	Selected pattern fills no byte within the selected VC-12 tributary payload during the measurement time interval.
Yellow	Bad data, such as would occur if the input cable from the BT OSU to the central monitoring facility were disconnected.

The traffic displayed in Fig 13 was generated by the ANDO tester that transmitted this traffic on to the SDH ring immediately before the BT OSU (see Fig 2). During the SDH EPMA feasibility demonstration, an ANRITSU traffic generator was attached to a spur off an ADM located remotely from the BT OSU. Even though the BT OSU could not perform pointer processing at the VC-12 level, the BT OSU with the EPMA successfully demonstrated both utilisation computation and pattern detection for the ANRITSU generated traffic.

Figure 14 shows the central monitoring facility display for the patterns detected from the ANRITSU generated traffic. Unlike the ANDO traffic generator, the ANRITSU traffic generator could place traffic on to only one tributary at a time. The other three tributaries passed random network data. This accounts for the green bars in this graph.

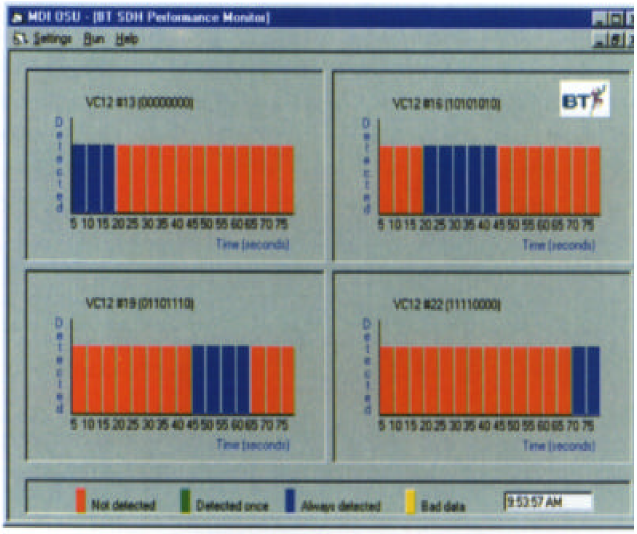

Fig 13 EPMA pattern detection (actual test data using ANDO traffic generator).

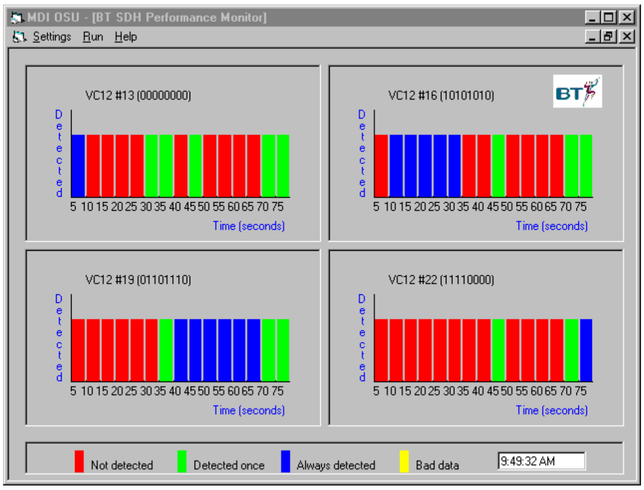

Fig 14 EPMA pattern detection (actual test data using ANRITSU

4. Summary and suggestions for future work

4.1 Summary

This paper has presented an enhanced performance measurement architecture (EPMA) that provides flexible, centralised, and distributed real-time information monitoring, analysis, and interpretation to support SDH network management. This paper has described how existing SDH interfaces, test tools, and network analysis devices can be used in support of and integrated into the EPMA, and it has suggested modifications to existing SDH network devices to provide basic EPMA functionality. The results of an SDH EPMA feasibility demonstration that took place in March 1996 have also been presented.

This demonstration successfully illustrated the basic EPMA functions using both existing and EPMA-customised SDH equipment on a model SDH network. The SDH EPMA has introduced a new central monitoring facility that uses a new information interpretation system to monitor and interpret performance information in SDH networks.

The SDH EPMA has also introduced new performance measures for SDH utilisation computation, and it has demonstrated the ability to monitor multiple functions simultaneously from a single work station. These functions include SDH network synchronisation accuracy monitoring, error performance and alarm monitoring, and utilisation computation and pattern detection at the VC-12 tributary level. The results presented here demonstrate the utility of the EPMA.

4.2 Suggestions for future work

The work described here can be extended for both SDH and ATM networks. For SDH networks the BT OSU could be modified to include pointer processing for VC-12 tributary boundaries and to include the additional registers needed to recognise additional patterns and all 63 VC-12 tributaries within the VC-4. For example, pseudo-random patterns as long as an entire STM-1 frame could possibly be detected and displayed. The central monitoring facility's graphical user interface could be enhanced to provide a uniform display of all functions monitored with the EPMA utilisation graphical interface serving as the standard. The information interpretation system could be enhanced to provide more detailed information concerning SDH network utilisation and other monitored functions. The information interpretation system could evolve into a true expert system in which the interpreted information feeds directly into a network management system that performs any necessary actions to remedy network problems. Monitoring ATM over SDH networks could also be investigated.

ATM may be a better place for the implementation of the EPMA architecture than SDH. For example, the EPMA architecture could provide insight into unused capacity in ATM networks by monitoring available bit rate (ABR) and providing information interpretation. Network operators could use this interpreted information to guarantee customer service but to delay dedicating bandwidth to that customer until it is needed.

An ATM local area network (LAN) performance monitoring feasibility demonstration, similar to that for the SDH EPMA, also could be conducted. Such a demonstration would apply EPMA to distributed, system level network management of ATM LANs.

4.3 Suggested applications of the SDH EPMA

The EPMA's ability to determine excess capacity on an SDH network could be used to alleviate excess demand for bandwidth from future BT customers. For example, BT could use the EPMA information interpretation system to suggest bandwidth reduction and corresponding cost savings for an existing customer with persistent excess capacity, and BT could offer service to future customers sooner using this extra bandwidth.

Another possible application for the EPMA information interpretation system could be to ensure that an equipped tributary is carrying actual traffic. For example, following an initial installation, a field engineer could accidentally leave a VC-12 tributary equipped, even though this channel was never connected to a customer.

BT field engineers and technicians could use a laptop PC equipped with EPMA reconfigurable hardware and software to provide a diagnostic tool to verify SDH network compliance for holdover and qualification. Such an EPMA facility could be reconfigured to verify that SDH network synchronisation performance meets specifications. Another EPMA reconfiguration within the same laptop PC could interpret information provided by a GPS satellite receiver to provide time interval error (TIE) testing information for SDH networks. By taking advantage of the memory on the EPMA-equipped laptop PC, the field engineer or technician could collect the network performance information and then bring it back to the laboratory for analysis.

The EPMA could also provide the field service team with the option of monitoring network activity remotely. Remote monitoring of customer networks permits the field service team to quickly identify the severity of a network problem and then to determine whether this problem jeopardises customer quality of service. If the problem warrants a site visit, the service team could respond immediately. If the problem were determined to be minor, then the service team could have the option of fixing it at the most economical time, such as during a planned visit to that particular site rather than late at night or on a weekend when the service rates are high.

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